

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.Tech Degree S8 (S) Examination September 2025 (2019 Scheme)

Course Code: CXT402**Course Name: ADVANCED COMPUTER GRAPHICS****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

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| 1 | What is depth cueing? | (3) |
| 2 | Explain the concepts of convex hulls and control graphs in simple terms? | (3) |
| 3 | Explain in detail about uvn viewing-coordinate reference frame. | (3) |
| 4 | Explain viewing pipeline with the help of a figure. | (3) |
| 5 | What do you mean by space-subdivision methods? Explain its types. | (3) |
| 6 | What do you mean by transparency in graphics? | (3) |
| 7 | Define any three basic components of a light. | (3) |
| 8 | Convert RGB value (255,128,64) into CMY. | (3) |
| 9 | What is zero acceleration? | (3) |
| 10 | Discuss different animation functions. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | a) List out the tests that could be performed by a graphics package. | (4) |
| | b) What are the basic concepts of polygon surface representation in computer graphics? Mention its advantages. | (10) |

OR

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|----|--|-----|
| 12 | a) Explain parametric and geometric continuity conditions in detail? | (8) |
| | b) Show that two curves $n(t)=(t^2+2t-2,t^2)$ and $r(t)=(t^2+2t+1,t+1)$ are both C^0 and G^0 continued where they join at $n(1)=r(0)$. Do they meet C^1 and G^1 continuity? | (6) |

Module II

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|----|---|-----|
| 13 | a) Perform Rotation transformation over a cube 'OABCDEFG' and rotate it through 45° in the anticlockwise direction about the y-axis. A(0,4,0), B(0,4,4), C(4,4,0), D(4,4,4), E(4,0,0), F(0,0,4), G(4,0,4), O(0,0,0). | (8) |
| | b) Discuss in detail about 3D viewing parameters. | (6) |

OR

- 14 a) Consider a cube "OABCDEFG" is given O(0, 0, 0), A(0, 4, 0), B(0, 4, 4), C(4, 4, 0), D(4, 4, 4), E(4, 0, 0), F(0, 0, 4), G(4, 0, 4) and we are given with Scaling factor $S_x=2$, $S_y=3$, $S_z=2$. Perform Scaling operation over the cube. (8)
- b) Explain in detail about oblique projection. Also mention its advantages. (6)

Module III

- 15 a) Describe basic illumination models with equations. (14)

OR

- 16 a) Explain different polygon rendering methods. (14)

Module IV

- 17 a) Explain CIE chromaticity diagram in detail. (8)
- b) Explain the algorithm for converting RGB color parameters to HSV values. (6)

OR

- 18 a) Explain in detail about different types of color models. (14)

Module V

- 19 a) Explain different types of interpolation used in computer graphics. (6)
- b) What are the ways in which the motions of objects can be specified in an animation system? Explain in detail. (8)

OR

- 20 a) Discuss different sequence used in animation. (8)
- b) Describe in detail about types of raster animation along with its application. (6)
